A Geostatistical Analysis of Soil Organic Carbon Distribution in Bavaria, Germany

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Abstract

Mapping Soil Organic Carbon (SOC) is vital for climate mitigation, yet Bavaria's complex topography makes accurate spatial prediction a challenge. This study evaluated the performance of deterministic (IDW), geostatistical (OK, EBK), and hybrid (EBKR) models using 118 topsoil samples from the LUCAS 2018 dataset. Exploratory Spatial Data Analysis confirmed a significant North-South gradient, with carbon hotspots concentrated in the Alpine foreland. While Ordinary Kriging achieved a low RMSE (16.45) after outlier removal, it produced an overly smoothed surface. The Empirical Bayesian Kriging Regression (EBKR) model, integrating Elevation and Nitrogen as covariates, proved to be the most robust approach. It effectively addressed landscape-scale non-stationarity, yielding the most geomorphologically realistic prediction surface and the highest local precision (minimum standard error of 1.88). The findings demonstrate that incorporating environmental auxiliary data is essential for regional carbon inventories, providing a precise baseline for sustainable land management and climate reporting in heterogeneous terrains.

Keywords: Bavaria, geostatistical analysis, interpolation, soil organic carbon.

1. Introduction

Soil organic carbon (SOC) represents the largest terrestrial carbon pool, containing significantly more carbon than the atmosphere and global vegetation combined (Chen et al., 2023; Y. Wang et al., 2011). Maintaining and increasing SOC stocks is essential for soil ecosystem services, including nutrient cycling, water retention, and the prevention of land degradation (Mustafa et al., 2020).

In Germany, mapping SOC at regional scales is essential for national greenhouse gas inventories and environmental reporting (Broeg et al., 2024). The distribution of SOC in Bavaria is highly heterogeneous, influenced by a complex interaction of climate, parent material, topography, and centuries of varied land-use history (Broeg et al., 2024).

Soil research is still reliant on traditional point-based sampling, but it does not provide the continuous coverage required for spatial decision-making (Sahu et al., 2021). Geostatistical methods, such as Ordinary Kriging (OK) and Empirical Bayesian Kriging (EBK), address this by modelling the spatial autocorrelation of SOC to predict values at unsampled locations (John et al., 2021; Veronesi & Schillaci, 2019). These methods provide the Best Linear Unbiased Estimate (BLUE) but often assume spatial stationarity, which may not hold true across a heterogeneous landscape like Bavaria (Sahu et al., 2021; Tziolas et al., 2024).

To enhance mapping accuracy, researchers have recently adopted Digital Soil Mapping (DSM) frameworks that utilize environmental covariates (Gibson et al., 2021; Y. Wang et al., 2011). Methods like Empirical Bayesian Kriging Regression (EBKR) allow for the integration of explanatory variables such as elevation and slope derived from Digital Elevation Models (DEMs), as well as soil physicochemical properties like Nitrogen and Clay content, to model the non-spatial variance of SOC (John et al., 2021; Sahu et al., 2021; Y. Wang et al., 2011).

Hence, the main objectives of this study were to:

- Assess the predictive accuracy of deterministic (IDW) and stochastic (OK and EBK) methods in mapping the spatial distribution of Soil Organic Carbon (SOC) in Bavaria.
- Investigate the integration of environmental covariates, including a terrain derivative (elevation) and a soil physicochemical property (Nitrogen) using the Empirical Bayesian Kriging Regression technique.
- Produce a definitive SOC prediction surface and an associated Prediction Standard Error map to quantify the spatial distribution of carbon stocks in Bavaria and identify high-uncertainty areas.

By determining the most accurate model for predicting SOC distribution, this study contributes to a more precise understanding of Bavaria's carbon stocks, supporting sustainable land management and climate change mitigation strategies.

2. Study Area and Data

2.1. Study Area

The study area considered is Bavaria, which is in the southeast of Germany, as shown in Figure 1. It has an area of 70,550km and is the largest state in Germany by land area. The study area falls between latitudes 47° 6′ - 50° 34′ N and longitudes 8° 58′ - 13° 51′ E.
Bavaria is characterized by significant environmental and topographic diversity, which can be divided into four major landscapes: the Northern Bavarian Triassic hill country, the Bavarian Forest, the Alpine Foreland, and the Limestone Alps in the extreme south (Wiesmeier et al., 2014).
This region lies in the warm-temperate zone, with a mean annual temperature of 8.6 °C and a mean annual precipitation of 892 mm. Elevations range from approximately 100m in the northwest to nearly 3,000m in the high-alpine regions. The predominant soils are Luvisols, Gleysols, Leptosols, Stagnosols, and Cambisols (Zepp et al., 2023).

2.2. Dataset

The primary dataset used in this project is the LUCAS 2018 Topsoil Data obtained from the European Soil Data Centre (ESDAC). It contained soil properties such as pH, organic carbon content, phosphorus, potassium, and nitrogen, covering the European Union and the United Kingdom (Fernandez-Ugalde et al., 2022). This dataset originally consisted of 18,984 samples, taken from 0-20 cm depth.
However, for the purpose of this project, the obtained dataset was clipped to the Bavaria boundary shapefile, obtained from the Database of Global Administrative Areas (GADM). This resulted in 124 sample points as shown in Figure 1, and 27 attributes,

including the points' geographic coordinates and elevation, SOC values and other soil properties.
Also, as a covariate for EBK Regression, an SRTM Digital Elevation Model (DEM) for the study area was obtained from the European Environment Agency.

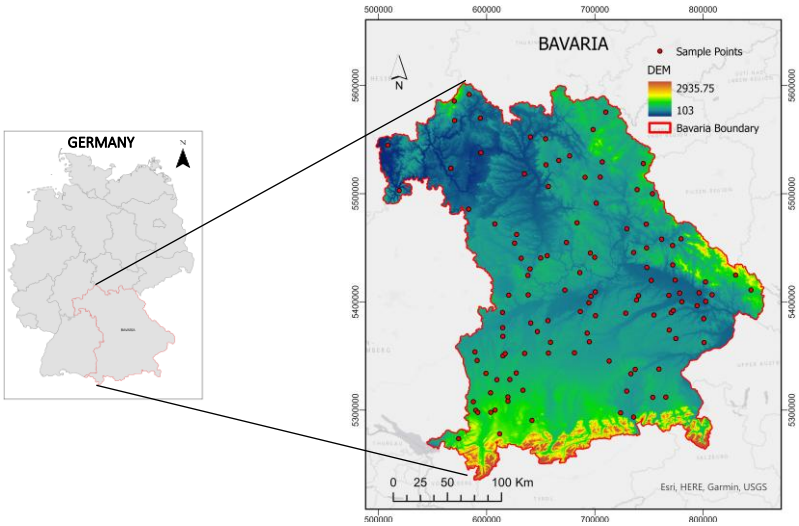

Figure 1: Distribution of topsoil samples in Bavaria, Germany

3. Methodology

3.1. Data Preprocessing

The initial dataset consisted of 124 soil records with 27 associated attributes. A rigorous data cleaning and dimensionality reduction workflow was implemented within ArcGIS Pro and Excel to ensure a robust geostatistical analysis. The data were first spatially clipped to the Bavarian state boundary to focus the study area.
Columns with a substantial number (80% or more) of missing values were eliminated to preserve the integrity of the model and avoid computational errors in the geostatistical wizard.
The final processed dataset comprised 118 high-quality records and 12 attributes, providing a clean foundation for the subsequent geostatistical interpolation methods and regression-kriging analysis.

3.2. Exploratory Data Analysis

A comprehensive Exploratory Data Analysis was conducted to characterise the distribution of SOC and its relationship with environmental drivers. Descriptive statistics, including the mean, median, and standard deviation (SD), were calculated in Excel and provided baseline measures of central tendency and variance, highlighting the significant heterogeneity of carbon stocks across Bavaria (Abdulmanov et al., 2021; Sahu et al., 2021). Also, a bivariate analysis was done to examine the correlation between the SOC values and their corresponding elevation (Z. Wang et al., 2025).

3.3. Exploratory Spatial Data Analysis (ESDA)

The data obtained from ESDAC was originally prepared in the WGS 1984 geographic coordinate system but had to be reprojected to the ETRS 1989 UTM Zone 32N coordinate system to guarantee consistency and enable additional spatial analysis. A data posting was done on ArcGIS Pro to visualize the spatial arrangement of sampling points. To assess

data distribution and stationarity, histograms, regional histograms, and Voronoi maps were generated (Abdulmanov et al., 2021; Sahu et al., 2021). Indicator maps were utilized to identify threshold-specific spatial patterns. Spatial autocorrelation was quantified using Global Moran's I, while Local Moran's I (Anselin) and outlier analysis identified "High-Low" and "Low-High" clusters (Saputra et al., 2023). Finally, focal and block statistics were applied to analyze neighbourhood-scale variability and examine whether a proportional effect might exist in the SOC data (Costa, 2024).

3.4. Spatial Interpolation

A variety of interpolation methods were employed to identify the optimal spatial predictor for SOC in Bavaria. Inverse Distance Weighting (IDW) served as the deterministic interpolator, assuming that points closer to one another are more similar than those further apart (Sahu et al., 2021). For geostatistical modelling, Ordinary Kriging (OK) was utilized to provide the Best Linear Unbiased Estimate (BLUE) by manually fitting a global semivariogram (John et al., 2021). Empirical Bayesian Kriging (EBK) was further applied to automate variogram modelling and account for local uncertainty (Sahu et al., 2021). Finally, EBK Regression integrated elevation and nitrogen as environmental covariates to address landscape-scale non-stationarity (Abdulmanov et al., 2021; John et al., 2021).

3.5. Model Validation and Accuracy Assessment

To evaluate model performance, a cross-validation framework was implemented using Mean Error (ME) and Root Mean Square Error (RMSE) as primary validation metrics. ME was used to detect systematic prediction bias, while RMSE quantified overall model accuracy and precision (Wiesmeier et al., 2014; Yuan et al., 2021). Final model selection was based on a comparative performance approach, identifying the interpolator that minimized error and most effectively captured the spatial structure of SOC across the Bavarian landscape.

4. Results

4.1. Descriptive Statistics

The content of SOC, which is measured in grammes of carbon per kilogramme of soil (g.kg-1), showed a median of 28.70 g.kg-1 and a mean of 32.80 g.kg-1 (Table 1). A positively skewed distribution is indicated by the difference between the mean and the median. It implies that whereas most sampling sites have a moderate carbon content, some localities have much greater amounts. With values ranging from a minimum of 8.10 g kg-1 to a maximum of 135.20 g.kg-1, the dataset demonstrated significant geographic heterogeneity. A standard deviation (SD) of 20.98 g.kg-1 further quantifies this significant variability, suggesting that a straightforward global mean is insufficient to describe the intricate distribution of carbon reserves throughout the Bavarian terrain.

Elevation, the environmental covariate utilized in regression modelling, varied from 245 to 891 meters, with a mean of 465.07 meters and a standard deviation of 129.42 meters. The varied physical features of the study area, which will probably influence the spatial distribution of SOC, are confirmed by this topographic variance.

Table 1: Descriptive statistics of SOC and the elevation variables.

Variable	Mean	Median	SD	Minimum	Maximum	Range	Sum
SOC (g kg^{-1})	32.80	28.70	20.98	8.10	135.20	127.10	3808.60
Elev (m)	465.07	454.70	129.42	245	891	646	54878

4.2. Exploratory Spatial Data Analysis

4.2.1. Data Posting

Figure 2 illustrates the spatial layout of the 118 topsoil samples, which shows a clear non-random distribution of carbon stocks throughout Bavaria. A visual examination reveals a clear North-South gradient. The southern and southwestern Alpine regions exhibit a notable clustering of high SOC values (> 45.3 g.kg-1), whereas the northern and central agricultural plains are dominated by lower to moderate SOC levels (< 29.3g.kg-1). The frequency distribution histogram, which shows a strong positive (right) skew, further illustrates this regional variability. The mean (32.8 g.kg-1) drifts to the right of the median (28.7 g.kg-1) as a result of the high-value outliers' tail extending considerably towards the maximum of 135.2 g.kg-1. This demonstrates that the dataset does not follow a normal distribution.

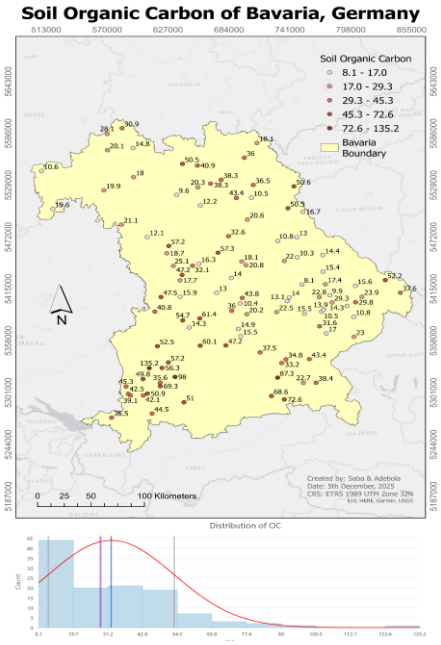

Figure 2: Data Posting and Distribution Histogram

4.2.2. Regional Histogram

To verify the dataset's spatial partitioning, regional histograms were used. The largest carbon stores are found in the Alpine regions, as shown in Figure 3, where choosing the southern sampling locations shows the upper tail of the histogram. On the other hand, samples from the northern and eastern regions only match the lower value ranges. The assumption of a constant mean throughout the study area is called into question by this clear separation, which supports the existence of first-order non-stationarity (Abdulmanov et al., 2021).

4.2.3. Indicator Maps

Quartile-based thresholds were used to create indicator maps that show the spatial probability of exceeding particular SOC concentrations. The spatial range of high-value places significantly decreases as one moves from the first quartile (Q_1) to the third quartile (Q_3), as illustrated in Figure 3. Although a wide range of sampling points reach the lower thresholds, only a small and clearly grouped number of locations surpass the Q_3 criterion (43.8 g.kg-1). The positive skewness observed in the descriptive statistics is visually reinforced by the lack of high-value sites.

4.2.4. Voronoi Maps

To study local stationarity, voronoi tessellations were built using mean and median statistics. In contrast to the northern and central agricultural landscapes, where

concentrations fall within the lowest classes (Mean: 15.7–23.4 g/kg; Median: 13.1–18.1 g/kg), the southern Alpine foreland is dominated by high-value polygons (Mean: 52.1–75.0 g/kg; Median: 47.5–68.6 g/kg), as shown in Figure 4. This strong gradient confirms that there is first-order non-stationarity and that the local mean of SOC is driven by a deterministic north-south trend rather than being constant throughout the research area.

4.2.5. Global and Local Moran's I statistic

Global Moran's I was computed to measure the level of spatial reliance in the SOC dataset. The analysis produced a positive Moran's Index of 0.21, along with a z-score of 3.24 and a p-value of 0.0012, as shown in Figure 5. The null hypothesis of random distribution can be rejected because these metrics are more than the critical threshold ($z > 1.96$, $p < 0.05$). The findings show that SOC values are spatially clustered rather than randomly distributed throughout Bavaria, confirming a mild but statistically significant positive spatial autocorrelation.

To identify these spatial instabilities, cluster and outlier analysis (Anselin Local Moran's I) was then carried out, as illustrated in Figure 5. The spatial patterning identified 54 features that were statistically significant. While 25 Low-Low clusters were focused in the central-east, 16 High-High clusters verified the southwest Alpine region as a carbon hotspot. Thirteen spatial outliers were also found, five of which were High-Low and eight of which were Low-High. These outliers indicate specific areas where SOC values differ significantly from those of their close neighbours.

4.2.6. Moving Window Statistics

To evaluate neighborhood-scale variability and check for stationarity, focal and block statistics were used. The focal mean maps, as illustrated in Figure 6, support the north-south gradient, with the southern and southwestern regions clearly dominated by elevated SOC concentrations. Importantly, Figure 7's focal standard deviation maps show that local variability is not constant, peaking in the southern regions where the mean SOC is also high. The existence of a proportional influence is suggested by this spatial relationship.

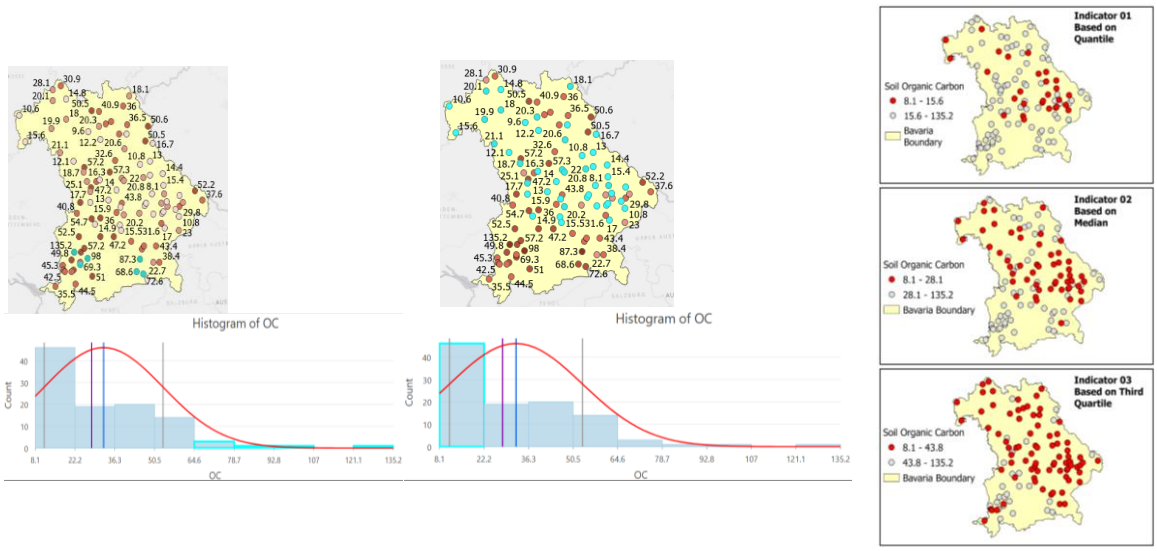

Figure 3: Regional Histograms(left) and Indicator maps based on Q1 - Q3(right).

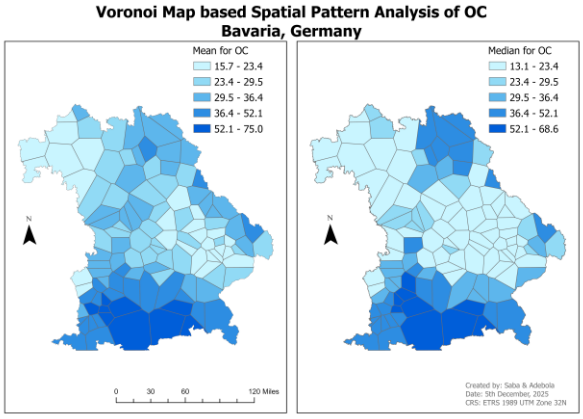

Figure 4: Voronoi Maps

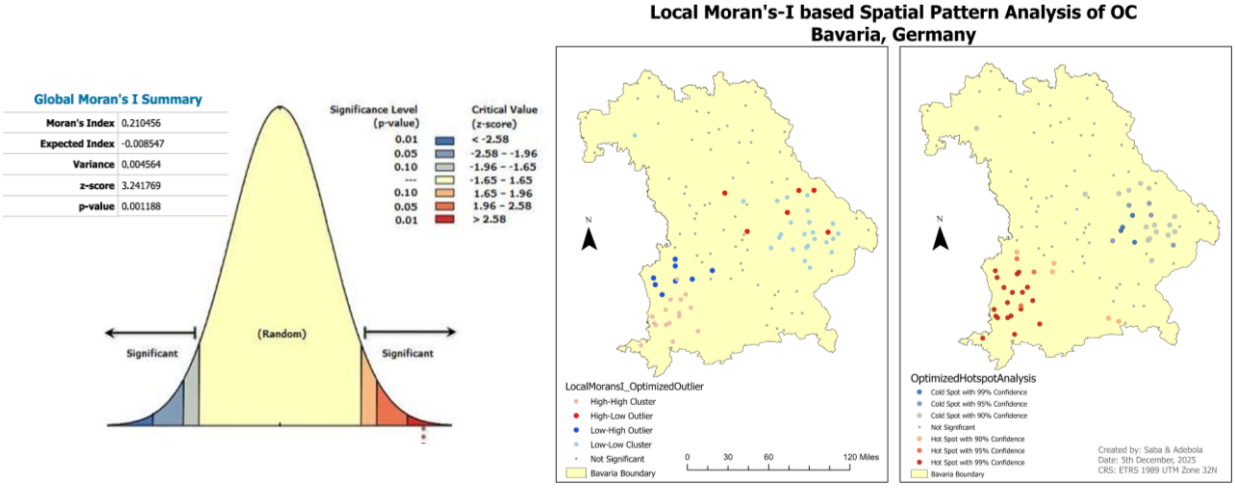

Figure 5: Global Morans I (left) and Local Moran's I (right)

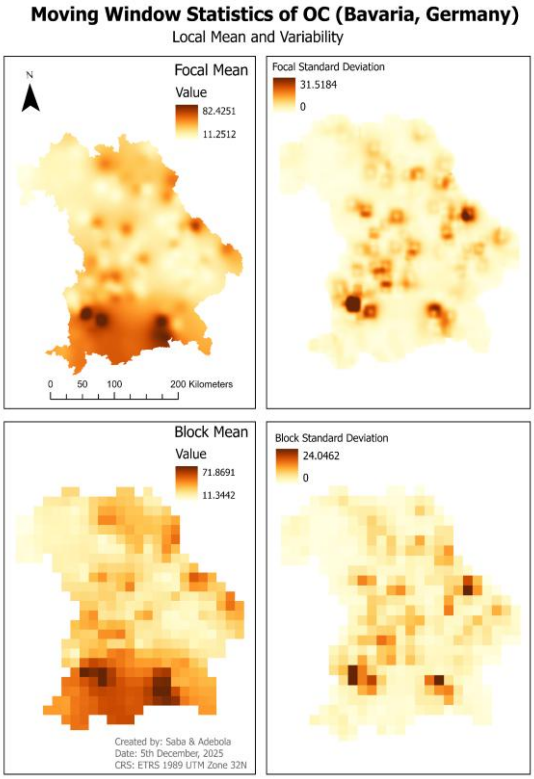

Figure 6: Focal and Block statistics Map of Bavaria Germany

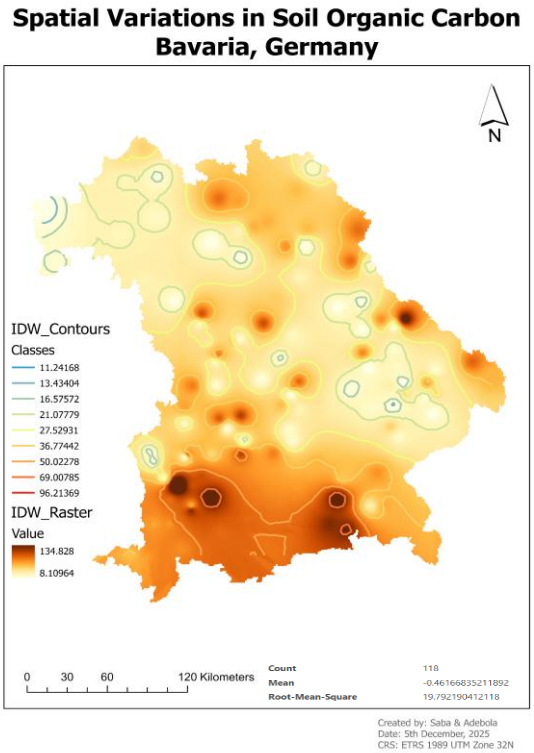

Figure 7: Inverse Distance Weighting predicted SOC surface

4.3. Interpolation Models

Four interpolation techniques were applied, and various parameter combinations were tried to get their resulting surfaces are discussed as follows. Appendix A shows the parameter combination tables for all techniques, while Appendix B shows the simulated semivariogram for all the Kriging models used.

4.3.1. Inverse Distance Weighting (IDW)

The Inverse Distance Weighting (IDW) approach was used as the deterministic baseline for spatial prediction. To determine the optimal search neighbourhood and weighting criteria, a thorough optimization process was carried out. Different combinations of sector types and neighbour counts were evaluated, as shown in Appendix A (Table A1). A Circle 8 sector neighbourhood with a minimum of 5 neighbours and a maximum of 8 neighbours was found to be the ideal arrangement, producing the lowest Mean Error (ME) of -0.4616 and Root Mean Square Error (RMSE) of 19.7921.

The broad regional trends, such as the increased carbon stores in the southern Alpine region, are effectively highlighted by the resulting prediction surface, which is displayed in Figure 7. But the image shows a distinctive "bullseye effect" surrounding isolated sampling locations, especially in the eastern area close to the European Green Belt. This visual artefact shows that IDW tends to exaggerate extremes in regions with sparse sampling and is quite sensitive to local data density (Yuan et al., 2021).

4.3.2. Ordinary Kriging (OK)

The Best Linear Unbiased Estimate (BLUE) of the SOC distribution was produced using Ordinary Kriging (OK). Local spatial outliers produced a "pure nugget effect," masking the underlying spatial structure, according to preliminary modelling. This was fixed by eliminating the 13 statistically significant outliers found by Local Moran's I, which significantly enhanced the semivariogram's structure (raising the partial sill from 0.09 to 0.202) and decreased the Root Mean Square Error (RMSE) from 18.861 to 16.449. As indicated in Table 2, the final model also demonstrated a lower Mean Error (ME) of 0.04, indicating a notable decrease in systematic bias. Its simulated semivariogram shown in Figure B1 (Appendix), has a nugget (0.202) relative to the sill (0.420), indicating significant micro-scale variation.

As illustrated in Figure 8(top), the resulting prediction surface clearly delineates the north-south gradient, thereby capturing the substantial carbon stocks in the southern Alpine foreland that the earlier model had failed to resolve. Uncertainty is spatially variable, as seen by the corresponding Prediction Standard Error map. The error is reduced (lighter zones) in well-sampled areas and dramatically increases with distance from sampling places. The southeast's data-poor regions have the largest uncertainty values (up to 15.667), indicating that although OK generates a statistically sound surface following the elimination of outliers, sample density still strongly limits its local reliability.

4.3.3. Empirical Bayesian Kriging (EBK)

Empirical Bayesian Kriging (EBK), which uses a spectrum of semivariograms as shown in Figure B1 (Appendix), to account for parameter uncertainty rather than fitting a single

deterministic model, was used to address local non-stationarity. The model's inherent resilience to spatial outliers was a crucial discovery. When applied to the entire dataset, EBK produced findings that were more geomorphologically plausible than Ordinary Kriging, with a Root Mean Square Error (RMSE) of 18.607 and a negligible Mean Error (ME) of -0.005. The seamless movement of carbon stocks from the northern plains to the southern highlands is well captured by the resultant spatially continuous prediction surface (Figure 8(middle)). Removing outliers significantly reduced visual quality even if the RMSE was numerically reduced to 16.618. This indicates that EBK's internal subsetting process adequately handles local abnormalities without the need for manual data exclusion. Its variogram model was Gaussian with a nugget of about 0.45 and a sill of 0.950.

The accuracy of these estimates is measured by the EBK Prediction Standard Error surface. The error is geographically dependent, similar to Ordinary Kriging, with the intensively sampled centre regions having the lowest uncertainty values (< 2.6). However, because there are few sampling points in the high-elevation zones, the error rises dramatically towards the study area boundaries, especially in the deep south and southeast, where uncertainty peaks at 20.5. The EBK model is robust in its trend estimation, but its local precision is still strongly constrained by the availability of ground-truth data in the outlying Alpine regions, as this sharp gradient in standard error shows.

4.3.4. Empirical Bayesian Kriging Regression (EBKR)

To improve geographical prediction, Empirical Bayesian Kriging Regression (EBKR) was used, including elevation (DEM) and nitrogen as covariates to account for the physical causes of SOC. By striking a balance between low prediction error and strong spatial realism, this hybrid strategy outperformed the EBK model and attained the study's most dependable performance, yielding a Root Mean Square Error (RMSE) of 16.68 and a Mean Error (ME) of 0.008. The resultant surface has a very complex texture, as seen in Figure 8 (bottom). As previously noted in the research by Gibson et al. (2021) and John et al. (2021), this accurately depicts the severe geographic gradients of carbon stocks in the southern Alpine region, showing that local terrain and nutrient factors greatly regulate SOC distribution. Also, its semivariogram showed a distinct shift in structure (with nugget of about 0.90 and sill of 0.980) because it models the residuals after accounting for Elevation and Nitrogen.

The enhanced reliability of the model is validated by the Prediction Standard Error surface. The regression model achieved a reduced minimal standard error (1.88) compared to traditional EBK (2.60). It showed much higher confidence in well-sampled locations, even though uncertainty naturally remained larger in the data-sparse southern highlands. The best statistically reliable estimate of Bavarian soil organic carbon is produced by integrating environmental covariates, which successfully fills in information gaps, as evidenced by the general decrease in global error.

Table 2: Cross-validation metrics of the four interpolation models.

Parameter	**IDW**	**OK**	**EBK**	**EBKR**
Mean Error (ME)	-0.461	0.040	-0.005	0.008
RMSE	19.792	16.449	16.618	16.680

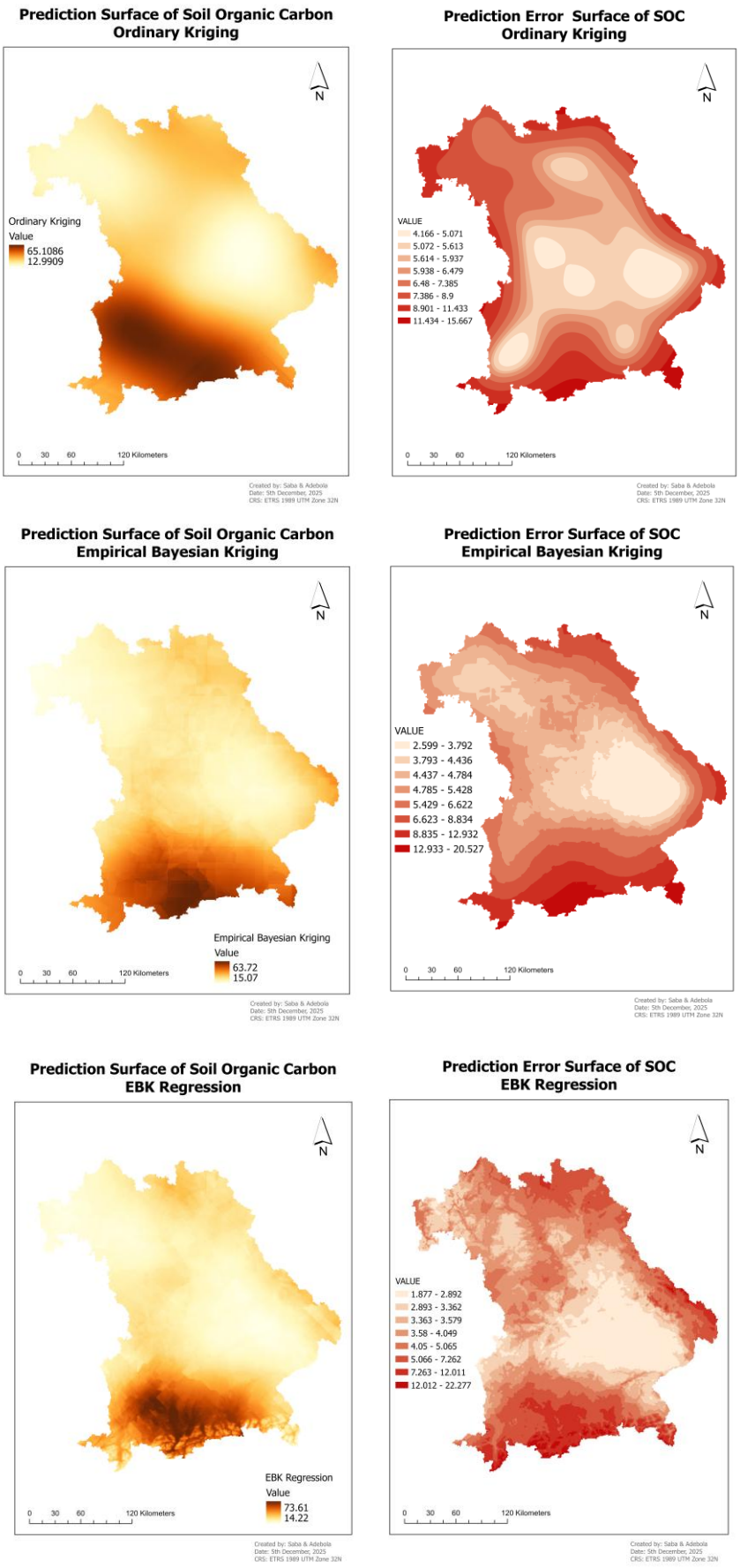

Figure 84: Final Prediction Surface Map and Error map for Ordinary Kriging(top), Empirical Bayesian Kriging(middle) and EBK Regression(bottom).

4.4. *Comparative Analysis*

4.4.1. *Model Divergence*

A difference surface was created by subtracting the Empirical Bayesian Kriging predictions from the Ordinary Kriging predictions in order to spatially quantify the differences between the two leading models. The map shows a sharp difference in model behaviour, as seen in Figure 9(left). The difference values are small (light hues) in the northern and central plains, suggesting that both models function similarly in homogeneous, low-variance areas.

The southern Alpine foreland, however, shows a notable divergence, with the difference peaking at 30.61 g/kg. The high-carbon clusters found in the ESDA exactly match this deep red zone. The significant positive difference demonstrates that the Ordinary Kriging model consistently underestimated the carbon reserves in this crucial area due to excessive smoothing. On the other hand, without compromising global stability, the EBK model effectively maintained these high values, offering a more accurate depiction of the localised carbon hotspots.

4.4.2. *Impact of Covariates*

A difference surface was created by contrasting the normal EBK model with the EBK Regression model to measure the precise contribution of environmental factors. Elevation has little predictive power in homogeneous terrain, as seen in Figure 9(right), where the divergence is low throughout the flat northern plains.

However, the difference is significantly greater in the southern Alpine region, ranging from -21.33 to 35.92 g/kg. The distinctive, ridge-like texture of the high-difference zones (dark red) perfectly complements the terrain of the area. This demonstrates that the local extremes were not captured by the typical EBK model, which successfully "smoothed over" the rough terrain. On the other hand, the fine-scale variability caused by elevation was successfully resolved by the EBK Regression model, indicating that accurate prediction in the complex Alpine landscape requires the integration of DEM and nitrogen data.

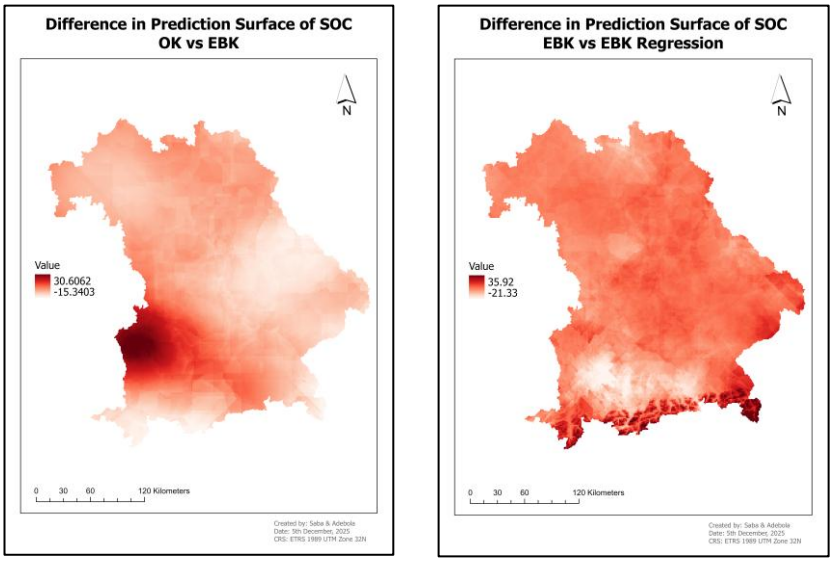

Figure 9: Comparative Analysis Maps, OK vs EBK(left) and EBK vs EBK Regression(right).

5. Conclusion

This study compared deterministic and geostatistical approaches for mapping Soil Organic Carbon (SOC) throughout the varied landscape of Bavaria. A large north-south gradient with a notable clustering of high values in the Alpine foreland was confirmed by exploratory spatial data analysis (ESDA), suggesting first-order non-stationarity that contradicts conventional stationarity assumptions.

Clear trade-offs between robustness and simplicity were found via comparative analysis. In areas with sparse data, the deterministic Inverse Distance Weighting (IDW) approach underestimated variance and produced "bullseye" artefacts. Ordinary Kriging (OK) was also very sensitive to local anomalies; although eliminating outliers improved the error numerically (RMSE: 16.449), the result was an excessively smoothed surface that masked important southern carbon hotspots.

Empirical Bayesian Kriging (EBK), on the other hand, produced a geomorphologically credible surface utilizing the entire dataset (RMSE: 18.607), exhibiting inherent robustness to non-stationarity. However, Empirical Bayesian Kriging Regression (EBKR) produced the final model. This hybrid approach effectively simulated the physical determinants of carbon buildup by using elevation and nitrogen as environmental factors. It produced the most accurate topographic soil gradient delineation and the lowest prediction error (RMSE: 16.68).

Finally, this study reveals that hybrid techniques that include environmental auxiliary data are critical for effectively mapping soil properties in difficult terrains. In order to support focused actions for sustainable land use and climate change mitigation measures in Bavaria, the developed prediction surface offers a crucial baseline for regional carbon inventory management.

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Appendix A

Final combinations are highlighted in RED

Table A1: IDW trials for cross-validation.

Sector Type	Maximum neighbors	Minimum neighbors	Mean Error	Root-Mean-Square error (RMSE)
Circle 1 sector	15	10	-0.5930	20.1157
Circle 1 sector	13	8	-0.5019	20.2222
Circle 1 sector	10	5	-0.5395	20.3120
Circle 1 sector	22	18	-0.6931	19.9864
Circle 1 sector	5	3	-0.3235	21.0412
Circle 1 sector	13	7	-0.5283	20.2135
Circle 1 sector	10	7	-0.5475	20.3126
Circle 4 sector	15	10	-0.5339	19.8234
Circle 4 sector	13	8	-0.5142	19.8199
Circle 4 sector	8	5	-0.5234	19.9300
Circle 4 sector	12	8	-0.5031	19.8380
Circle 4 sector	9	5	-0.5207	19.9240
Circle 4 sector	25	15	-0.5244	19.8278
Circle 4 sector	20	15	-0.5451	19.8258
Circle 4 sector	17	14	-0.5366	19.8214
Circle 4 sector	22	18	-0.5602	19.8161
Circle 4 sector	9	6	-0.5397	19.9274
Circle 4 sector	5	3	-0.5872	19.9272
Circle 4 sector	13	7	-0.5441	19.8260
Circle 4 sector	10	7	-0.5148	19.9052
Circle 8 sector	25	15	-0.4909	19.8504
Circle 8 sector	22	18	-0.4919	19.8417
Circle 8 sector	20	15	-0.5032	19.8260
Circle 8 sector	12	8	-0.4869	19.8142
Circle 8 sector	9	6	-0.4720	19.8033
Circle 8 sector	8	5	-0.4616	19.7921
Circle 8 sector	7	4	-0.4182	19.8322
Circle 8 sector	5	3	-0.4071	19.8874
Circle 4 sector, 45° offset	25	15	-0.4844	19.8459
Circle 4 sector, 45° offset	22	18	-0.4851	19.8218
Circle 4 sector, 45° offset	20	15	-0.4934	19.8049
Circle 4 sector, 45° offset	17	14	-0.4806	19.8048
Circle 4 sector, 45° offset	15	10	-0.4865	19.8112
Circle 4 sector, 45° offset	13	7	-0.4546	19.8372
Circle 4 sector, 45° offset	10	7	-0.4075	19.8528
Circle 4 sector, 45° offset	12	8	-0.4334	19.8420
Circle 4 sector, 45° offset	10	5	-0.4075	19.8528
Circle 4 sector, 45° offset	9	6	-0.3777	19.9009
Circle 4 sector, 45° offset	8	5	-0.3833	19.915
Circle 4 sector, 45° offset	7	4	-0.4245	19.9415
Circle 4 sector, 45° offset	5	3	-0.3669	20.0778

Table A2: Trials for Ordinary Kriging

Sector Type	Maximum neighbors	Minimum neighbors	Mean Error	Root-Mean-Square error (RMSE)
Circle 1 sector	15	10	-0.0893	16.4425
Circle 1 sector	13	8	-0.1297	16.5365
Circle 1 sector	10	5	-0.2330	16.7177
Circle 1 sector	8	5	-0.1304	16.7202
Circle 1 sector	12	8	-0.1596	16.5956
Circle 1 sector	15	4	-0.0793	16.4478
Circle 1 sector	6	4	0.1394	17.3787
Circle 1 sector	22	18	0.0859	16.4932
Circle 1 sector	17	14	-0.0089	16.5243
Circle 1 sector	9	6	-0.2641	16.6798
Circle 1 sector	5	2	0.16763	16.0900
Circle 1 sector	13	7	-0.1441	18.3137
Circle 1 sector	10	7	-0.0831	18.2870
Circle 4 sector	15	10	-0.1443	17.9117
Circle 4 sector	13	8	-0.1488	17.9361
Circle 4 sector	8	5	-0.1933	17.9509
Circle 4 sector	12	8	-0.1387	17.9435
Circle 4 sector	9	5	-0.1535	17.9641
Circle 4 sector	25	15	-0.1880	17.8784
Circle 4 sector	20	15	-0.1925	17.9073
Circle 4 sector	17	14	-0.1819	17.9078
Circle 4 sector	22	18	-0.2136	17.8790
Circle 4 sector	9	6	-0.1598	17.9700
Circle 4 sector	5	2	-0.2174	17.7766
Circle 4 sector	13	7	-0.1302	17.9387
Circle 4 sector	10	7	-0.1197	17.9933
Circle 8 sector	25	15	-0.1805	17.8312
Circle 8 sector	22	18	-0.1796	17.8766
Circle 8 sector	20	15	-0.1820	17.8309
Circle 8 sector	17	14	-0.1793	17.8527
Circle 8 sector	15	10	-0.1656	17.8581
Circle 8 sector	13	7	-0.1126	17.8804
Circle 8 sector	10	7	-0.1124	17.8999
Circle 8 sector	10	5	-0.1071	17.9030
Circle 8 sector	12	8	-0.1323	17.8665
Circle 8 sector	9	6	-0.0681	17.9195
Circle 8 sector	8	5	-0.0876	17.9151
Circle 8 sector	7	4	-0.0497	17.9366
Circle 8 sector	5	3	0.0841	17.9523
Circle 4 sector, 45° offset	25	15	-0.0147	17.8824
Circle 4 sector, 45° offset	22	18	0.025	16.8815
Circle 4 sector, 45° offset	20	15	0.099	16.8725
Circle 4 sector, 45° offset	17	14	0.0388	16.449
Circle 4 sector, 45° offset	15	10	0.086	16.8845
Circle 4 sector, 45° offset	13	7	-0.102	16.9049

Table A3: Trials for Empirical Bayesian Kriging.

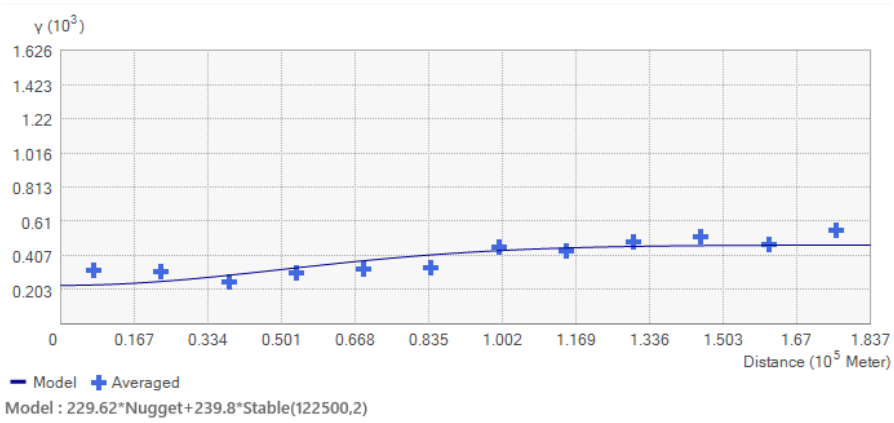
Parameter	**Parameter Value**	**Max Neighbors**	**Min Neighbors**	**Average CRPS**	**Inside 95% CI**	**ME**	**RMSE**	**ME_STD**	**RMSE_STD**	**ASE**
Subset size	200									
Overlap factor	2									
No. Simulations	300	15	10	9.8775	97.4576	-0.0311	18.9554	-0.0228	1.0847	18.3339
Transformation	Log Emp.									
Sector Type	1 sector									
Subset size	200									
Overlap factor	2									
No. Simulations	300	15	10	9.8205	98.3050	0.1515	18.8310	-0.0025	1.0723	18.1575
Transformation	Log Empirical									
Sector Type	1 sector									
Subset size	200									
Overlap factor	2									
No. Simulations	300	25	15	9.7523	97.4576	-0.0507	18.7513	-0.0064	1.0692	17.9916
Transformation	Log Empirical									
Sector Type	1 sectors									
Subset size	200									
Overlap factor	2									
No. Simulations	300	20	10	9.8369	97.4576	0.0313	18.8904	-0.0067	1.0782	18.0699
Transformation	Log Empirical									
Sector Type	1 sectors									
Subset size	200									
Overlap factor	2									
No. Simulations	300	8	3	9.7617	97.4576	0.0217	18.7234	-0.0036	1.0647	18.0242
Transformation	Log Empirical									
Sector Type	4 sectors									
Subset size	200									
Overlap factor	2									
No. Simulations	300	8	**4**	9.6748	97.4576	-0.0058	18.6074	0.0013	1.0505	17.9754
Transformation	Log Emp.									
Sector Type	4 sectors									
Subset size	200									
Overlap factor	2									
No. Simulations	300	16	7	5.75	93.61	-0.07	10.44	-0.00	0.99	10.54
Transformation	Log Empirical									
Sector Type	4 sectors									

Table A4: Trials for Empirical Bayesian Kriging Regression.

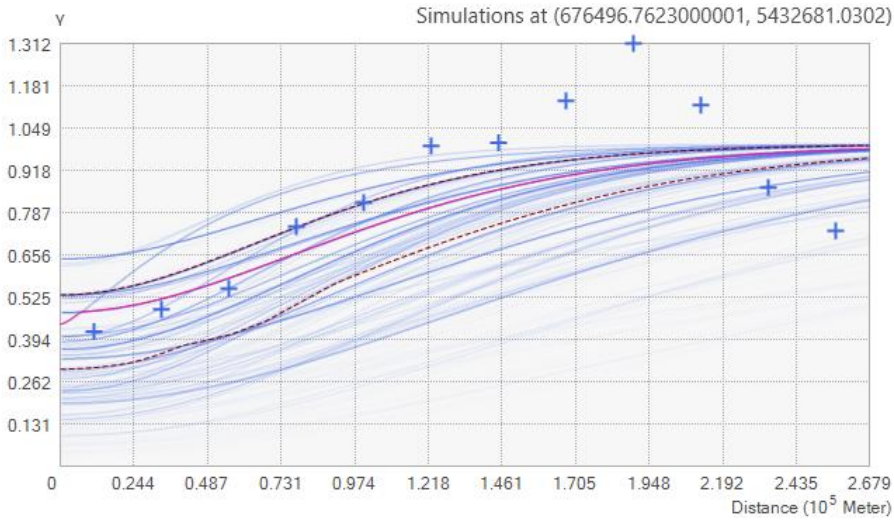
Parameter	**Parameter Value**	**Max Neighbors**	**Min Neighbors**	**Average CRPS**	**Inside 95% CI**	**ME**	**RMSE**	**ME_STD**	**RMSE_STD**	**ASE**
Subset size	100									
Overlap factor	1									
No. Simulations	100	15	10	8.7015	97.4576	-0.0989	16.6853	-0.0055	0.9767	17.3685
Transformation	Log Emp.									
Sector Type	1 sector									
Subset size	100									
Overlap factor	1									
No. Simulations	100	17	15	8.6443	97.4576	-0.1475	18.8310	-0.0033	0.9686	17.2836
Transformation	Log Emp.									
Sector Type	1 sector									
Subset size	100									
Overlap factor	1									
No. Simulations	100	25	18	8.5936	97.4576	-0.1749	16.4766	-0.0014	0.9610	17.2480
Transformation	Log Emp.									
Sector Type	1 sector									
Subset size	100									
Overlap factor	1									
No. Simulations	100	13	8	8.7371	97.4576	-0.0673	16.7334	-0.0063	0.9798	17.4158
Transformation	Log Emp.									
Sector Type	1 sector									
Subset size	100									
Overlap factor	1									
No. Simulations	100	8	5	8.7267	97.4576	0.1665	16.7072	0.0070	0.9624	17.6235
Transformation	Log Emp.									
Sector Type	1 sector									
Subset size	100									
Overlap factor	1									
No. Simulations	100	**10**	**6**	**8.715**	**97.4576**	**0.0088**	**16.6832**	**-0.0012**	**0.9664**	**17.4950**
Transformation	Log Emp.									
Sector Type	1 sector									
Subset size	100									
Overlap factor	1									
No. Simulations	100	7	4	8.7760	97.4576	0.2353	16.8067	0.0081	0.9678	17.7002
Transformation	Log Emp.									
Sector Type	1 sector									

Appendix B

a



b



c

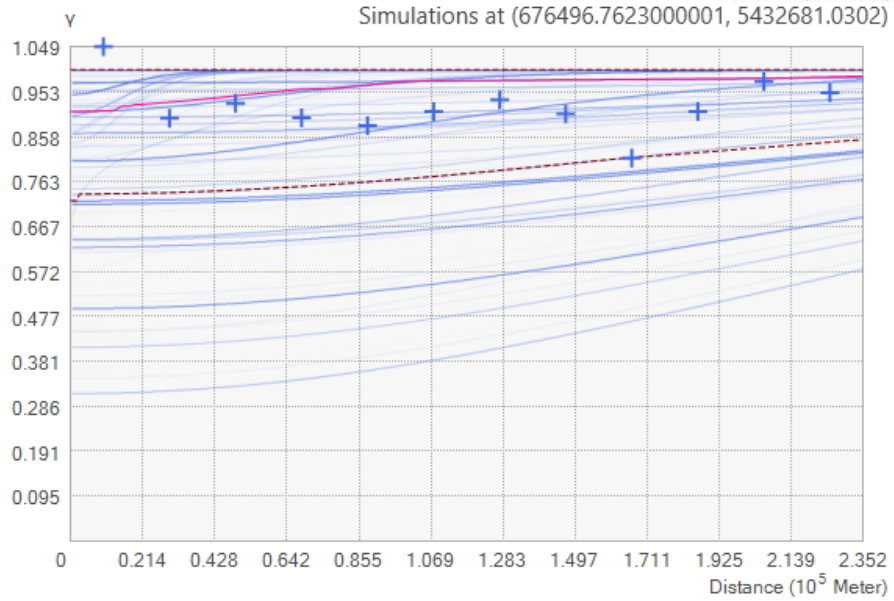


Figure B1: Simulated semivariograms for (a) Ordinary Kriging, (b) Empirical Bayesian Kriging and (c) Empirical Bayesian Kriging Regression